Vemre HRIS

Enterprise Human Resource Intelligence System

Comprehensive Product & User Documentation

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Prepared for VemreHR Enterprise

Developer and Web App Company Logos

Overview

What the platform delivers

Vemre HRIS is an enterprise-grade Human Resource Intelligence System that unifies employee data, performance management, payroll, attendance, onboarding, knowledge, and communication into a single, secure platform. It is designed for growing organizations that need structured HR workflows, fast reporting, and an excellent employee experience.

Why it is valuable

- Centralizes HR data and processes to reduce manual work.
- Enables leadership visibility with dashboards and performance analytics.
- Automates payroll, attendance, and KPI scoring with audit trails.
- Improves employee engagement with self-service, chat, and knowledge tools.

Core Features

Modules that power Vemre HRIS

Feature Highlights

- Employee management, profiles, documents, and roles
- Performance KPIs, templates, scoring, and summaries
- Payroll runs, salary structures, and payslip generation
- Attendance tracking with geofencing and late policy logic
- Leave requests, approvals, and HR ticketing
- Recruitment job postings and applicant workflow
- Knowledge base with versioning and search
- Onboarding guides and progress tracking
- Internal chat with read receipts, replies, and attachments
- System notifications for new chat messages (desktop and mobile)

Technology

Stack and architecture

Frontend: React + Vite, Tailwind CSS, React Router, Axios
Backend: Django 5, Django REST Framework, SimpleJWT
Mobile: Capacitor (Android build)
Storage: SQLite (local dev), PostgreSQL recommended for production

High-level flow

1. React frontend calls API endpoints via JWT
2. Django API enforces permissions and business logic
3. Data stored in database and files served via media endpoints
4. Notifications dispatched for chat events

Installation - Backend

Step-by-step setup

- Navigate to backend directory.
- Create and activate a virtual environment.
- Install dependencies from requirements.txt.
- Run migrations and create an admin user.
- Start the Django development server.

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python manage.py migrate
python create_fresh_admin.py
python manage.py runserver 0.0.0.0:8000
```

Installation - Frontend & Mobile

Web and Android

- Navigate to frontend directory.
- Install dependencies with npm.
- Set VITE_API_URL to your backend API base (e.g., http://localhost:8000/api).
- Run the dev server for web.

```
cd frontend
npm install
npm run dev
# Android build
npm run build
npx cap sync android
npx cap open android
```

Usage & Screenshots

Key touchpoints

1. Sign in with admin or staff credentials.
2. Use the dashboard to track KPIs, payroll, attendance, and insights.
3. Open the chat module to collaborate with colleagues and receive notifications.
4. Staff can access self-service pages for profile, leaves, expenses, and documents.

Splash Screen

Login Screen

Screenshot placeholder - replace with live capture

Careers Page

Screenshot placeholder - replace with live capture

Chat Module

Screenshot placeholder - replace with live capture

Chat Notifications

System alerts for new messages

Vemre HRIS now triggers system notifications when a new chat message arrives. On desktop web, the browser Notification API is used. On Android builds, Capacitor Local Notifications surface native alerts.

- Users are prompted once for permission.
- Notifications appear when the app is not in focus.
- Unread counts are monitored for new inbound messages.
- Pluggable for future push delivery (FCM) when app is closed.

Troubleshooting & Roadmap

Keep the platform running smoothly

Login network error

Ensure VITE_API_URL points to the backend API base (example: http://localhost:8000/api). The backend must be reachable on the same network and port.

Recommended improvements

- Add real-time messaging via WebSockets for instant chat updates.
- Enable true push notifications with Firebase Cloud Messaging.
- Introduce role-based dashboards with configurable widgets.
- Switch to PostgreSQL for production scalability.
- Add audit logging exports and SOC-ready reporting.